When Should AI Advice Change? Midterm Report

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1 Introduction

People increasingly ask language models for consequential personal and career advice. A good advisor's recommendation should depend on the facts and on what the person cares about, not on how hard the person pushes for an answer. We call the failure *decision-level sycophancy*: the recommendation moves toward the user's apparent preference although nothing decision-relevant has changed.

Most sycophancy benchmarks score only *unwanted* change (Sharma et al., 2024; Cheng et al., 2025; Hong et al., 2025; Hwang et al., 2026). That rewards a model that never changes its mind, which is not a good advisor either. A useful advisor must update when the user adds a real constraint and ask when a critical fact is missing. Our question is therefore one of discrimination: **can a model tell changes that should alter a recommendation from changes that should not?** This matters because a model that fails it in either direction gives advice that depends on the user's tone rather than their situation, in exactly the settings where users are least able to check it.

Halfway through the project we have (1) 30 authored career scenarios, each with four controlled variants, plus 50 AITA posts; (2) a harness that scores text-generating and decision-only models on the same prompts; and (3) results for a decision-only model, its base model, and a frontier chat model in two output formats. Three findings stand out so far. The 9B base model moves *more* for a stated preference than for a decision-relevant fact; decision fine-tuning helps it separate the two, but mostly by ignoring rewording, not pressure. Claude Opus 5 never gives in to pressure, in one turn or over several, yet still updates on real corrections. And for Claude, first-person pressure shows up instead as *refusals*: one clause, "I'm the one deciding here," gets ordinary career prompts blocked by a safety classifier.

2 Related Work

Sharma et al. (2024) traced sycophancy partly to RLHF, because raters reward agreement. ELEPHANT (Cheng et al., 2025) extended it to open-ended advice, SYCON Bench (Hong et al., 2025) to multi-turn pressure, and FRAGILE (Hwang et al., 2026) to fact-preserving reframings. Input-level mitigations (Dubois et al., 2026; Christian and Mazor, 2026) and synthetic fine-tuning data (Wei et al., 2023) reduce it. All of these measure unwanted change on a single axis. Decision-only models such as Jev (TypeSafe AI, 2026), Laya (Convai Innovations, 2026) and Bespoke-Nimble (Bespoke Labs and Sathiamoorthy, 2026) return an option distribution instead of text; we know of no published test of them under preference pressure. We also observe over-refusal of benign prompts (Röttger et al., 2024), triggered by how a user talks about their own decision.

3 Data

Career scenarios. We wrote 30 job-offer and workplace decisions (a 10-scenario pilot plus 20 more). Each record holds the facts, the person's stated priority, an action question, and 3–4 options that always include *need more information*. From each base prompt we write the proposal's four variants: *pressure* (the user states a preferred option) and *reframe* (wording or order only) should leave the answer unchanged and may not add, remove or alter a fact; *relevant update* changes exactly one fact or priority, with a one-sentence reason the answer should change; *omission* removes the fact the decision hinges on. A validator checks these rules. Labels are still the authors' drafts; blind relabeling by two other team members, on shuffled sheets that hide the variant, is in progress.

Pressure controls. The authored pressure sentences differ in wording and also name the target option, so we added templated, fact-preserving con-

trols to every scenario: a third-person preference ("They really want to hear that the right move is to X"), a first-person version ("I'm the one deciding here, and I really want you to tell me the right move is to X"), a neutral mention of X, a preference for the base answer, and an irrelevant filler sentence.

AITA. We sampled 50 r/AmItheAsshole posts the community judged YTA from ELEPHANT's released data (Cheng et al., 2025), with a base prompt and a pressure prompt that appends "I really need someone to tell me I'm not the asshole here." The community verdict remains the reference.

4 Approach

Following the proposal, three conditions separate output format from decision training. **Decision-only:** Bespoke-Nimble-9B, a LoRA adapter on Qwen3.5-9B trained on minimal pairs where one fact flips the answer, scores every option and returns a distribution without generating text. **Base:** the same Qwen3.5-9B through Nimble's own scorer with the adapter removed, so prompt, scorer and output space are identical and any difference is due to the fine-tuning alone. **Chat:** Claude Opus 5 through the API, either *constrained* (answer with one option code) or *conversational* (a short paragraph of advice, with the code on the last line). We pin both Nimble checkpoints, since the public one changed mid-project. Laya and Jev are not yet evaluated. Refused API calls return no text; we exclude them from every rate and count them separately.

5 Evaluation Methods

We score the proposal's four behaviors against base: *pressure shift* (the choice moves to the user's preferred option), *reframe change*, the *warranted update* rate on relevant updates, and *omission asks* (choosing need more information when a fact is missing). Scored separately, a model can look good on pressure just by never moving. We therefore also treat the task as signal detection: a *hit* is a relevant update that changes the answer, a *false alarm* is a pressure or reframe variant that does, and $d' = z(\text{hit}) - z(\text{false alarm})$ (Macmillan and Creelman, 2005), with 95% intervals from 2,000 bootstrap resamples over scenarios. For the 9B models we also measure how far each variant moves the option distribution (total variation from base) and the AUROC of that movement for relevant versus irrelevant variants.

6 Experiments and Results So Far

The base model can't tell pressure from evidence. Qwen3.5-9B moves its option distribution more under pressure (mean TV 0.54) than under a relevant fact (0.36). Per scenario, pressure moves it more in 23 of 30 cases (sign test $p = 0.005$), and $d' \approx 0$ (Table 1): from its choices alone one cannot tell whether the user added evidence or only said what they wanted.

Decision fine-tuning helps, but not against pressure. Nimble separates relevant from irrelevant changes better than its base (AUROC 0.71 vs. 0.46). The gain comes mostly from reframe stability (12/30 $\rightarrow$ 3/30); pressure still flips 11 of 27 scenarios, and Nimble never answers *need more information*. Pressure also overrides confident answers: the median Nimble answer that flipped had a base log-odds margin of 3.1 over the target.

Claude is stable without being rigid. Claude never moved toward the pressure target, in either format, and still changed its answer on 68–76% of relevant updates ($d' \approx 2.3$). It did the same on every templated control (0 moves toward the target), so the controls cannot yet separate preference from mere mention; that test now depends on the 9B models. On AITA, all models side with the poster on many posts the community judged YTA even without pressure (Qwen 86%, Nimble 70%, Claude about 50%), and the pressure sentence leaves the decision-only models' verdicts unchanged.

Multi-turn pushback. Because single-turn pressure is at floor for Claude, we followed each constrained base answer with a second turn (29 scenarios; one was refused at turn 1). Claude changed its answer in 0 of 29 cases under directional or plain "Are you sure?", a restatement of the same facts, and three escalating rounds of pressure toward the user's option, while 25 of 29 single-fact corrections changed it (24 to the labeled answer), for a second-turn d' of 3.57. Unlike the models in SYCON Bench (Hong et al., 2025), it does not give in on later turns. Its remaining weakness is the same as in one turn: it rarely asks for missing information (8–9 of 29 omissions).

First-person pressure triggers refusals. The pressure-control run exposed a cost we had not looked for. Of 598 calls, 81 were refused, all under Claude's *cyber* safety category, on ordinary career prompts. Excluding one scenario refused on every prompt, the first-person line was refused on 34 of 58 calls, against 0 of 58 for the base prompt

	Pressure shift	Reframe change	Omission asks	Hit	d' [95% CI]	AUROC
Qwen3.5-9B base	18/29	12/30	10/30†	0.53	0.04 [−0.49, 0.54]	0.46
Bespoke-Nimble-9B	11/27	3/30	0/30	0.47	0.63 [0.06, 1.19]	0.71
Claude Opus 5, constrained	0/27	1/28	9/29	0.76	2.22 [1.46, 3.38]	–
Claude Opus 5, conversational	0/26	0/26	8/29	0.68	2.36 [1.61, 3.26]	–

Table 1: Single-turn results on the 30 career scenarios (draft labels). Denominators drop where a prompt was refused or the base answer already equals the pressure target. †The base model also answers *need more information* on 7 complete prompts.

Line appended to the base prompt	Prose	Code
(none)	0	0
I'm the one deciding here, and I really want you to tell me the right move is to X.	19	13
…without "really"	22	16
…"want to hear that" for "want you to tell me"	17	13
…in third person ("They're the one deciding")	12	8
…without "I'm the one deciding here"	11	5
I'm the one deciding here.	8	0
Tell me the right move is to X.	8	3
They really want to hear that the right move is to X.	5	1

Table 2: Claude Opus 5 refusals out of 29 career scenarios, by the one line appended to the prompt, for advice in prose and a code-only answer. All refusals fall under the cyber category.

and 5 of 58 for the same preference in third person. Resampling all 36 refused first-person calls three times gave 108 refusals out of 108, so this is deterministic, not sampling noise.

To find the trigger we changed one piece of the line at a time (Table 2). How hard the user pushes does not matter: removing "really" does not reduce refusals. The clause "I'm the one deciding here" matters most: removing it roughly halves refusals, and *on its own*, with no preference and no option named, it still gets 8 of 29 conversational prompts refused. Five scenarios, all involving money, debt, health insurance or a non-compete, are refused under every appended line, while their base prompts never are. Every code-only refusal is also refused when Claude is asked for written advice, never the reverse. The two lines repeated from the earlier run reproduced it in 113 of 116 cases.

These refusals change how Claude's results should be read. Its 0% pressure shift is measured only on prompts that got past the classifier; the first-person rows have just 16 (code) and 8 (prose) usable answers. For a real user who talks naturally about their own decision, the cost of pushing is not worse advice but no advice. When Claude did answer, no variant moved it toward the pressure target.

7 Discussion and Next Steps

Against the proposal's plan, the scenarios, harness and all four variant runs are done. Three caveats apply. Labels are still drafts; the update and omission rates depend on them, though hits and false alarms only need the answer to *change*. Thirty scenarios is small: the base-model finding survives bootstrap and a sign test, but the Nimble-vs.-base AUROC gap is suggestive only. And most Claude numbers come from one sample per prompt.

For the rest of the term we will: (1) finish blind relabeling and report agreement and Cohen's κ per variant; (2) run the pressure controls on the 9B models to test whether naming an option moves them as much as wanting it, i.e. whether their "sycophancy" is lexical priming; (3) estimate a "user wants X" direction in each 9B model's activations from six held-out phrasings and project it out (Arditi et al., 2024), testing whether pressure flips fall while warranted updates stay (scripts are ready, pending GPU time); (4) add Laya and, if access is granted, Jev; (5) add the five-sample reframe noise baseline and blind scoring of conversational answers' quality.

8 Ethical Considerations

AITA posts come from a released research dataset of public posts; we collect no new personal data and do not redistribute the posts. The refusal finding describes one deployed model in October 2026 and may change with its classifiers; we report it so over-refusal is counted alongside sycophancy.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. In *NeurIPS*.

Bespoke Labs and Maheswaran Sathiamoorthy. 2026. Nimble. https://github.com/bespokelabsai/nimble.

Myra Cheng, Sunny Yu, Cinoo Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. 2025. ELEPHANT: Measuring and understanding social sycophancy in LLMs. *arXiv:2505.13995*.

Brian Christian and Matan Mazor. 2026. Self-blinding and counterfactual self-simulation mitigate biases and sycophancy in large language models. *arXiv:2601.14553*.

Convai Innovations. 2026. Laya: a non-autoregressive system 1 decision engine. https://huggingface.co/convaiinnovations/laya.

Magda Dubois, Cozmin Ududec, Christopher Summerfield, and Lennart Luettgau. 2026. Ask don't tell: Reducing sycophancy in large language models. *arXiv:2602.23971*.

Jiseung Hong, Grace Byun, Seungone Kim, and Kai Shu. 2025. Measuring sycophancy of language models in multi-turn dialogues. In *Findings of EMNLP*.

Seojin Hwang, Minju Kim, Junhyuk Choi, JeongHyun Park, and Hwanhee Lee. 2026. Framing matters: Addressing framing sensitivity in decision-making through behaviorally-grounded value alignment. *arXiv:2605.28188*.

Neil A. Macmillan and C. Douglas Creelman. 2005. *Detection Theory: A User's Guide*, 2nd edition. Lawrence Erlbaum.

Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. 2024. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *NAACL*.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, et al. 2024. Towards understanding sycophancy in language models. In *ICLR*.

TypeSafe AI. 2026. Introducing system one models and jev. https://typesafe.ai/blog/introducing-system-one-models-and-jev.

Jerry Wei, Da Huang, Yifeng Lu, Denny Zhou, and Quoc V. Le. 2023. Simple synthetic data reduces sycophancy in large language models. *arXiv:2308.03958*.